TELIPORT
Script Your Future
Season 4

CASE STUDY 1: ACTG — PREDICTIVE MAINTENANCE (SELF-HEALING MACHINES)

FlairGuard AI: Connected Fleet Reliability Platform

Focus: Reliable Predictive-Maintenance Decisions Under Imperfect Telemetry | Team: FlairGuard | NASA C-MAPSS Benchmark

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1. End-to-End System Architecture

- Closed-Loop Pipeline: Streaming Ingestion -> Quality Audit -> Causal Features -> Dual ML Engine -> Deterministic Triage -> Persistent Work Orders.
- Telemetry Quality First: Evaluates physical sensor envelope, flatlining, and cycle monotonicity before allowing automated decisions.
- Sub-10ms Latency: Lightweight non-LLM decision logic ensures deterministic execution suitable for connected fleet operations.
- Verified vs Planned: Ingestion, causal features, triage, and SQLite store verified in Python (67 tests). FastAPI backend and operator dashboard planned for Round 2.

2. C-MAPSS Data & Causal Features

- Multi-Regime Coverage: Evaluated on NASA C-MAPSS FD001 (single condition) and FD002 (6 operating conditions across altitude/Mach).
- Strict Causal Features: Rolling stats (w=5, 20) computed strictly on past cycles ($1 \le \tau \le t$). Zero lookahead bias or data leakage.
- Regime Normalization: K-means operating condition clustering on flight settings with cluster-level standard scaling for FD002.
- Target Formulation: Piecewise linear degradation target capped at 125 cycles during training; official test evaluation remains uncapped.

3. ML Modeling & Champion Policy

- Model Benchmarking: Ridge, Random Forest, HistGB, and LightGBM evaluated via 5-fold group-disjoint cross-validation (zero engine leakage).
- Validated Champion: Tuned Point-RUL Policy remains our validated operational champion. Advanced validity gates are maintained as experimental.
- Empirical FD001 Score: LightGBM achieved official test MAE of 13.56 cycles, RMSE of 18.61 cycles, and NASA score of 606.02.
- Empirical FD002 Score: Regime-aware normalization improved FD002 test RMSE from 50.37 to 48.88 cycles under multi-operating conditions.

4. Deterministic Triage & Work Orders

- Explainable Decision Rules: Non-LLM deterministic policy evaluates RUL point predictions, uncertainty bands, and anomaly persistence.
- Actionable Severities: Critical ($RUL \le 15$ cyc), High ($RUL \le 35$ cyc, persistence ≥ 2), Medium ($RUL \le 55$ cyc), and Nominal ($RUL > 55$ cyc).
- Persistent SQLite Store: Idempotent work orders prevent alert storms via duplicate suppression; escalates severity as wear progresses.
- Full Audit Trail: Immutable logging of engine ID, cycle, severity, decision rationale, and raw telemetry snapshot for regulatory review.

5. Handling Imperfect Telemetry & Conflict

- Fault Separation: Explicitly separates data acquisition defects (stuck sensors, missing data, out-of-bounds values) from physical wear.
- Persistence Filtering: Configurable consecutive-cycle filter (K=3) reduces nominal early-life false alarms by 75% on C-MAPSS FD001.
- Evidence Conflict Escalation: When high anomaly persistence conflicts with optimistic RUL (> 60 cyc), ticket routes to human review.
- Unchecked Autonomy Guard: When sensor quality fails near maintenance thresholds, system refuses blind action and safely escalates (ESCALATE_HUMAN).

6. MLOps Lifecycle & Feasibility Evidence

- Verified Test Foundation: 67 passing unit and property tests verifying engine isolation, causal invariance, deduplication, and crash recovery.
- Continuous Learning: Retraining triggered on verified newly completed run-to-failure engines; promotion gated by validation criteria.
- Fault Isolation: Synthetic fault-injection sessions are strictly sandboxed from normal training data and operational model state.
- Scope Honesty: System models simulated degradation wear; does not claim physical engine repair or verified commercial cost savings.